

L^AT_EX Cheat Sheet

Document classes

`article` Best suited for exercise sheets.

Used at the very beginning of a document: `\documentclass{class}`. Use `\begin{document}` to start contents and `\end{document}` to end the document.

Common documentclass options

`10pt/11pt/12pt` Font size.
`letterpaper/a4paper` Paper size.
`twocolumn` Use two columns.
`twoside` Set margins for two-sided.
`landscape` Landscape orientation.
`draft` Double-space lines.
Usage: `\documentclass[opt,opt]{class}`.

Recommended packages

`multicol` Use n columns: `\begin{multicols}{n}`.
`mathtools` Advanced math typesetting. Includes `amsmath`.
`geometry` Page layout (margins, etc.)
`fancyhdr` Customize header and footer
`graphicx` Show image: `\includegraphics[width=x]{file}`.
`hyperref` Insert URL: `\url{https://...}`
or hyperlink: `\href{URL}{text}`.
`enumitem` Customize `enumerate`.
`tabularx` Stretchable X columns in tables.
`tabulary` Automatically balanced column widths.
`tabto` Tab stops (compatible with enumerations).
`beamer` For creating slides.
Use before `\begin{document}`.
Usage: `\usepackage{package}`

Title

`\author{text}` Author of document.
`\title{text}` Title of document.
`\date{text}` Date.

These commands go before `\begin{document}`. The declaration `\maketitle` goes at the top of the document.

Miscellaneous

`\pagestyle{empty}` Empty header, footer and no page numbers.
`\tableofcontents` Add a table of contents here.

Document structure

<code>\part{title}</code>	<code>\subsubsection{title}</code>
<code>\chapter{title}</code>	<code>\paragraph{title}</code>
<code>\section{title}</code>	<code>\subparagraph{title}</code>
<code>\subsection{title}</code>	

Use a $*$, as in `\section*{title}`, to not number a particular item—these items will also not appear in the table of contents.

Text environments

`\begin{comment}` Comment (not printed).
Requires `verbatim` package.
`\begin{quote}` Indented quotation block.
`\begin{quotation}` Like `quote` with indented paragraphs.
`\begin{verse}` Quotation block for verse.

Lists

`\begin{enumerate}` Numbered list.
`\begin{itemize}` Bulleted list.
`\begin{description}` Description list.
`\item text` Add an item.
`\item[x] text` Use x instead of normal bullet or number. Required for descriptions.

References

`\label{marker}` Set a marker for cross-reference, often of the form `\label{sec:item}`.
`\ref{marker}` Give section/body number of marker.
`\pageref{marker}` Give page number of marker.
`\footnote{text}` Print footnote at bottom of page.

Floating bodies

`\begin{table}[place]` Add numbered table.
`\begin{figure}[place]` Add numbered figure.
`\begin{equation}[place]` Add numbered equation.
`\caption{text}` Caption for the body.
The *place* is a list valid placements for the body. t =top, h =here, b =bottom, p =separate page, $!$ =place even if ugly. Captions and label markers should be within the environment.

Text properties

Font face

Command	Declaration	Effect
<code>\textrm{text}</code>	<code>\rmfamily text</code>	Roman family
<code>\textsf{text}</code>	<code>\sffamily text</code>	Sans serif family
<code>\texttt{text}</code>	<code>\ttfamily text</code>	Typewriter family
<code>\textmd{text}</code>	<code>\mdseries text</code>	Medium series
<code>\textbf{text}</code>	<code>\bfseries text</code>	Bold series
<code>\textup{text}</code>	<code>\upshape text</code>	Upright shape
<code>\textit{text}</code>	<code>\itshape text</code>	<i>Italic shape</i>
<code>\textsl{text}</code>	<code>\slshape text</code>	<i>Slanted shape</i>
<code>\textsc{text}</code>	<code>\scshape text</code>	SMALL CAPS SHAPE
<code>\emph{text}</code>	<code>\em text</code>	<i>Emphasized</i>
<code>\textnormal{text}</code>	<code>\normalfont text</code>	Document font
<code>\underline{text}</code>		<u>Underline</u>

The command `(ttt)` form handles spacing better than the declaration `(ttt)` form.

Font size

<code>\tiny</code>	tiny	<code>\Large</code>	Large
<code>\scriptsize</code>	scriptsize	<code>\LARGE</code>	LARGE
<code>\footnotesize</code>	footnotesize		
<code>\small</code>	small	<code>\huge</code>	huge
<code>\normalsize</code>	normalsize		
<code>\large</code>	large	<code>\Huge</code>	Huge

These are declarations and should be used in the form `{\small ...}`, or without braces to affect the entire document.

Verbatim text

`\begin{verbatim}` Verbatim environment.
`\begin{verbatim*}` Spaces are shown as `␣`.
`\verb!text!` Text between the delimiting characters (in this case ‘!’) is verbatim.

Justification

Environment	Declaration
<code>\begin{center}</code>	<code>\centering</code>
<code>\begin{flushleft}</code>	<code>\raggedright</code>
<code>\begin{flushright}</code>	<code>\raggedleft</code>

Better typesetting (by enabling hyphenation) can be achieved using the package `ragged2e` and its environments `RaggedRight`, `RaggedLeft`, and `Center`.

Miscellaneous

`\linespread{x}` changes the line spacing by the multiplier x .

Text-mode symbols

Symbols

<code>&</code>	<code>\&</code>	<code>_</code>	<code>_</code>	<code>...</code>	<code>\ldots</code>	<code>•</code>	<code>\textbullet</code>
<code>\$</code>	<code>\\$</code>	<code>^</code>	<code>\^{}{}</code>	<code> </code>	<code>\textbar</code>	<code>\</code>	<code>\textbackslash</code>
<code>%</code>	<code>\%</code>	<code>~</code>	<code>\~{}{}</code>	<code>#</code>	<code>\#</code>	<code>\$</code>	<code>\\$</code>

Delimiters

<code>`</code>	<code>“</code>	<code>”</code>	<code>{\}</code>	<code>[</code>	<code>(</code>	<code><</code>	<code>\textless</code>
<code>'</code>	<code>”</code>	<code>”</code>	<code>}\}</code>	<code>]</code>	<code>)</code>	<code>></code>	<code>\textgreater</code>

Dashes

Name	Source	Example	Usage
hyphen	<code>-</code>	X-ray	In words.
en-dash	<code>--</code>	1–5	Between numbers.
em-dash	<code>---</code>	Yes—or no?	Punctuation.

Line and page breaks

`\\` Begin new line without new paragraph.
`\\*` Prohibit pagebreak after linebreak.
`\kill` Don’t print current line.
`\pagebreak` Start new page.
`\columnbreak` Start new column.
`\noindent` Do not indent current line.

Spaces and rules

`~` Fixed space, disallow linebreak (N.~Biever).
`\smallskip` Small vertical space.
`\medskip` Medium vertical space.
`\bigskip` Big vertical space.
`\hspace{l}` Horizontal space of length l (Ex: $l = 20\text{pt}$).
`\vspace{l}` Vertical space of length l .
`\rule{w}{h}` Line of width w and height h .

Rules of width/height 0 are sometimes useful for solving spacing or alignment issues.

Tables and tab stops

Tab stops: tabbing vs. tabto

\= Set tab stop. \> Go to tab stop.
Tab stops can be set on “invisible” lines with \kill at the end of the line.
Normally \\ is used to separate lines.

The tabbing environment is incompatible with enumerate/itemize. In this case it’s better to use the package tabto instead:

\TabPositions{2cm, 5cm}
Some\tab sample\tab text.

tabular environment

\begin{array}[pos]{cols}
\begin{tabular}[pos]{cols}
\begin{tabular*}{width}[pos]{cols}

tabular column specification

l Left-justified column.
c Centered column.
r Right-justified column.
p{width} Same as \parbox[t]{width}.
@{decl} Insert decl instead of inter-column space.
| Inserts a vertical line between columns.

tabular elements

\hline Horizontal line between rows.
\cline{x-y} Horizontal line across columns x through y.
\multicolumn{n}{cols}{text}
 A cell that spans n columns, with cols column specification.

Math mode

For inline math, use \(\dots\) or \$...\$. For displayed math, use \[...\] or \begin{equation}.

2^x `2^{x}` x_0 `x_{0}`
 $\frac{x}{y}$ `\frac{x}{y}` $\sqrt[n]{x}$ `\sqrt[n]{x}`
 $\sum_{k=1}^n$ `\sum_{k=1}^n` $\prod_{k=1}^n$ `\prod_{k=1}^n`
 $\int_a^b f$ `\int_a^b f` $\iint_S$ `\iint_S`
 $\vec{u}$ `\vec{u}` $\overrightarrow{AB}$ `\overrightarrow{AB}`

Matrices and determinants:

`\begin{pmatrix}` $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
`1&2\\ 3&4`
`\end{pmatrix}`
`\begin{vmatrix}` $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$
`1&2\\ 3&4`
`\end{vmatrix}`

Math-mode symbols

$\leq$	<code>\leq</code>	$\geq$	<code>\geq</code>	$\neq$	<code>\neq</code>	$\approx$	<code>\approx</code>
$\times$	<code>\times</code>	$\div$	<code>\div</code>	$\pm$	<code>\pm</code>	$\cdot$	<code>\cdot</code>
$^\circ$	<code>\circ</code>	$\circ$	<code>\circ</code>	$\prime$	<code>\prime</code>	$\cdots$	<code>\cdots</code>
∞	<code>\infty</code>	$\neg$	<code>\neg</code>	$\wedge$	<code>\wedge</code>	$\vee$	<code>\vee</code>
$\supset$	<code>\supset</code>	$\forall$	<code>\forall</code>	$\in$	<code>\in</code>	$\rightarrow$	<code>\rightarrow</code>
$\subset$	<code>\subset</code>	$\exists$	<code>\exists</code>	$\notin$	<code>\notin</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\cup$	<code>\cup</code>	$\cap$	<code>\cap</code>	$ $	<code>\mid</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\dot{a}$	<code>\dot{a}</code>	$\hat{a}$	<code>\hat{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>
α	<code>\alpha</code>	β	<code>\beta</code>	γ	<code>\gamma</code>	δ	<code>\delta</code>
ϵ	<code>\epsilon</code>	ζ	<code>\zeta</code>	η	<code>\eta</code>	ε	<code>\varepsilon</code>
θ	<code>\theta</code>	ι	<code>\iota</code>	κ	<code>\kappa</code>	ϑ	<code>\vartheta</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	ν	<code>\nu</code>	ξ	<code>\xi</code>
π	<code>\pi</code>	ρ	<code>\rho</code>	σ	<code>\sigma</code>	τ	<code>\tau</code>
υ	<code>\upsilon</code>	ϕ	<code>\phi</code>	χ	<code>\chi</code>	ψ	<code>\psi</code>
ω	<code>\omega</code>	Γ	<code>\Gamma</code>	Δ	<code>\Delta</code>	Θ	<code>\Theta</code>
Λ	<code>\Lambda</code>	Ξ	<code>\Xi</code>	Π	<code>\Pi</code>	Σ	<code>\Sigma</code>
Υ	<code>\Upsilon</code>	Φ	<code>\Phi</code>	Ψ	<code>\Psi</code>	Ω	<code>\Omega</code>

Sample L^AT_EX document

```
\documentclass[11pt]{article}
\title{Template}
\author{Name}
\begin{document}
\maketitle

\section{section}
\subsection*{subsection without number}
text \textbf{bold text} text. Some math: $2+2=5$
\subsection{subsection}
text \emph{emphasized text} text. Display math:
\[
\int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}
\]
```

A centered table:

```
\begin{center}
\begin{tabular}{|l|c|r|}
\hline
first & row & data \\
second & row & data \\
\hline
\end{tabular}
\end{center}
\end{document}
```

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<https://wch.github.io/latexsheet/>
<https://github.com/gschintgen/latexsheet>